

Feedback — How Did Yeast Become a Wine Maker?

(Quiz 1)

[Help Center](#)

Thank you. Your submission for this quiz was received.

You submitted this quiz on **Fri 15 May 2015 1:44 PM PDT**. You got a score of **9.00** out of **10.00**. You can [attempt again](#), if you'd like.

Question 1

True or False: The fact that only 13% of genes in *S. cerevisiae* are duplicated does not rule out the possibility of a whole-genome duplication.

Your Answer		Score	Explanation
☒ True	✓	1.00	
☐ False			
Total		1.00 / 1.00	

Question 2

True or False: In practice, the number of clusters of biological data is typically not known in advance.

Your Answer		Score	Explanation
☒ True	✓	1.00	
☐ False			
Total		1.00 / 1.00	

Question 3

True or False: *MaxDistance* is less sensitive to outliers than the squared error distortion.

Your Answer	Score	Explanation
☐ True		
☒ False	✓ 1.00	
Total	1.00 / 1.00	

Question 4

How many ways are there to cluster 5 points into two clusters?

You entered:

15

Your Answer	Score	Explanation
15	✓ 2.00	
Total	2.00 / 2.00	

Question 5

True or False: The Lloyd algorithm is guaranteed to converge for any collection of data points and any choice of initial centers.

Your Answer	Score	Explanation
☐ True		
☒ False	✗ 0.00	
Total	0.00 / 1.00	

Question 6

Compute $MaxDistance(Data, Centers)$ for the following *Data* and *Centers*:

Data: (2, 8), (2, 5), (6, 9), (7, 5), (5, 2)

Centers: (3, 5), (5, 4)

You entered:

Your Answer		Score	Explanation
5	✓	1.00	
Total		1.00 / 1.00	

Question 7

Compute $Distortion(Data, Centers)$ for the following *Data* and *Centers*:

Data: (2, 8), (2, 5), (6, 9), (7, 5), (5, 2)

Centers: (3, 5), (5, 4)

You entered:

Your Answer		Score	Explanation
9	✓	1.00	
Total		1.00 / 1.00	

Question 8

Give the center of gravity of the following (three-dimensional) data points. Enter your answer in the form (x, y, z).

(17, 0, -4), (3, 14, 23), (9, 7, 16), (7, 3, 5)

You entered:

(9, 6, 10)

Your Answer		Score	Explanation
(9, 6, 10)	✓	2.00	
Total		2.00 / 2.00	